

Ex 2A: Shell Script

Aim:

To write a shell script to display a basic calculator.

Program:

```
#!/bin/bash

echo "Enter the first number: "
read a
echo "Enter the second number: "
read b
echo "1.Add 2.Subtract 3.Multiply 4.Divide"
read op
case $op in
1)echo "Result: $((a+b))";;
2)echo "Result: $((a-b))";;
3)echo "Result: $((a*b))";;
4)echo "Result: $((a/b))";;
*)echo "Invalid option";;
esac
```

Output:

```
[abhijith@abhijith shell]$ sh calculator.sh
enter two numbers:
10
5
add 15
sub 5
mul 50
div 2
mod 0
[abhijith@abhijith shell]$
```

Result:

Thus the program to execute a shell script to display a basic calculator has been executed successfully.

Ex 2B: Shell Script

Aim:

To write a shell script to check if a given year is a leap year or not.

Program:

```
#!/bin/bash

echo "Enter a year to check leap year: "
read a
if [ $((a%400)) -eq 0 ] || { [ $((a%4)) -eq 0 ] && [ $((a%100)) -ne 0 ]; }
then
echo "$a is a leap year"
else
echo "$a is not a leap year"
fi
```

Output:

```
[abhijith@abhijith shell]$ sh leap.sh
enter the year:
12
leap year
[abhijith@abhijith shell]$
```

Result:

Thus the program to execute a shell script to check if a given year is a leap year or not has been executed successfully.